

AIX

Robotics (2)

Mingyu Kim,
June, 5th, 2026

Part 1 — From LLMs to Robots

Physical AI and the New Robotics Thesis

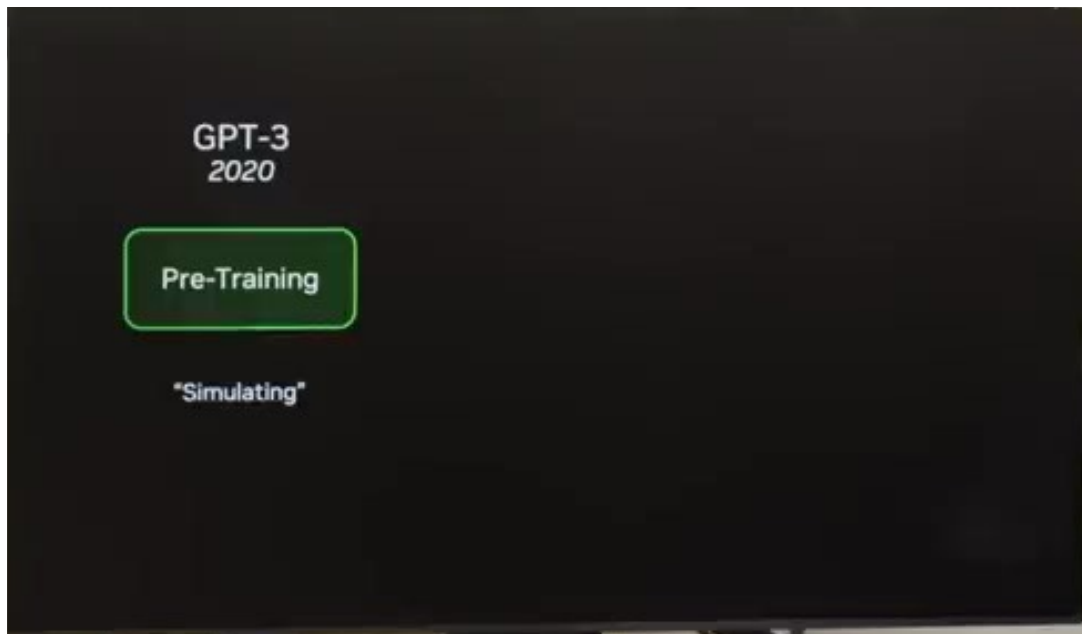
- Robotics as the next frontier after the LLM race reached a mature recipe.
- The key claim is that robotics may follow the same scaling logic that worked for language models.

DGX-1, OpenAI, and the Early Scaling Mindset

- The DGX-1 anecdote grounds the talk in the early compute era of modern deep learning.

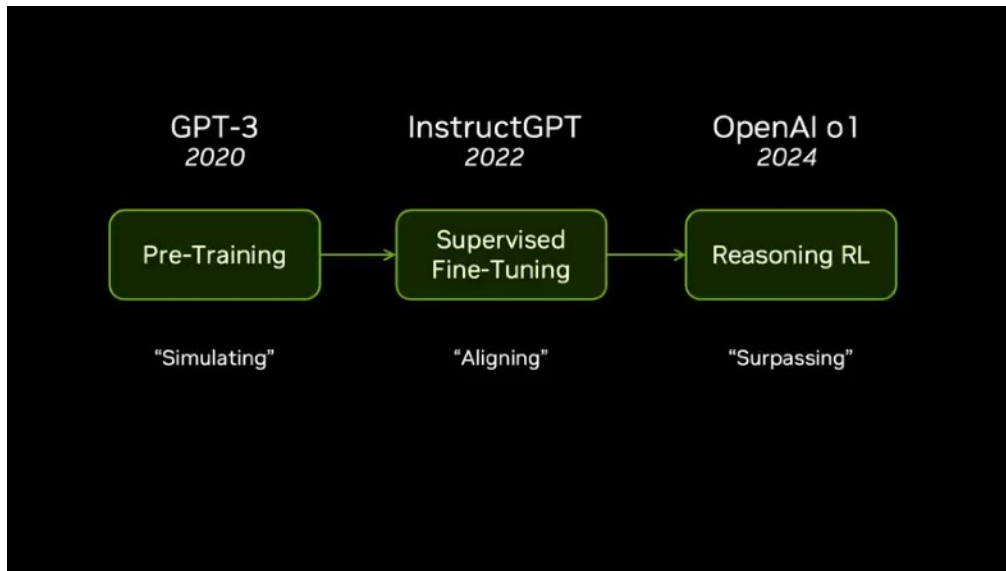
Pre-Training as Simulation

- Jim interprets GPT-style pre-training as simulation over internet-scale text.
- This phase produces a broad world model, but not yet instruction following.



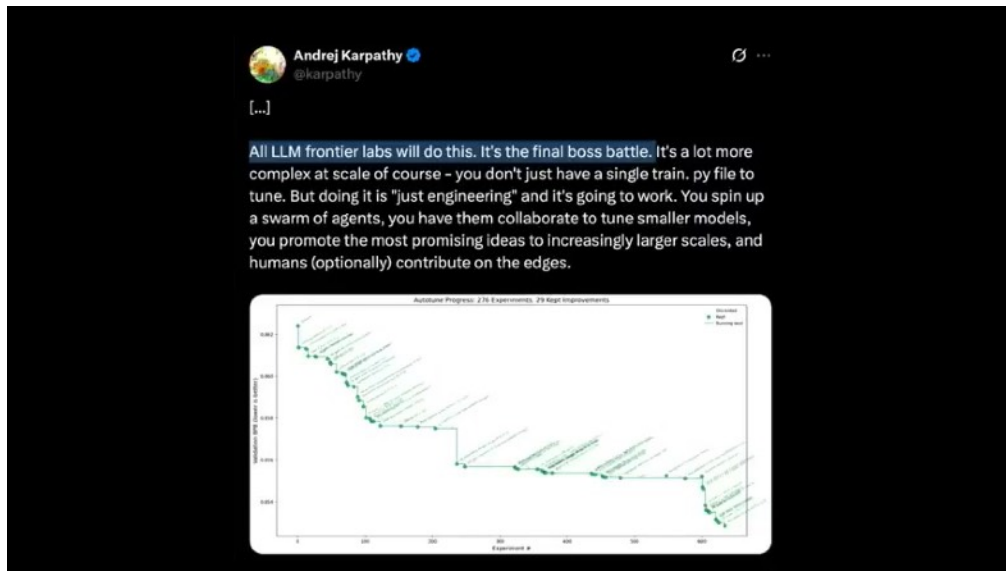
The Three-Phase Recipe for Frontier Models

- The canonical recipe is pre-training, supervised alignment, and reinforcement learning for surpassing.
- This structure becomes the conceptual template for the rest of the talk.



Reasoning RL as the Final Scaling Frontier

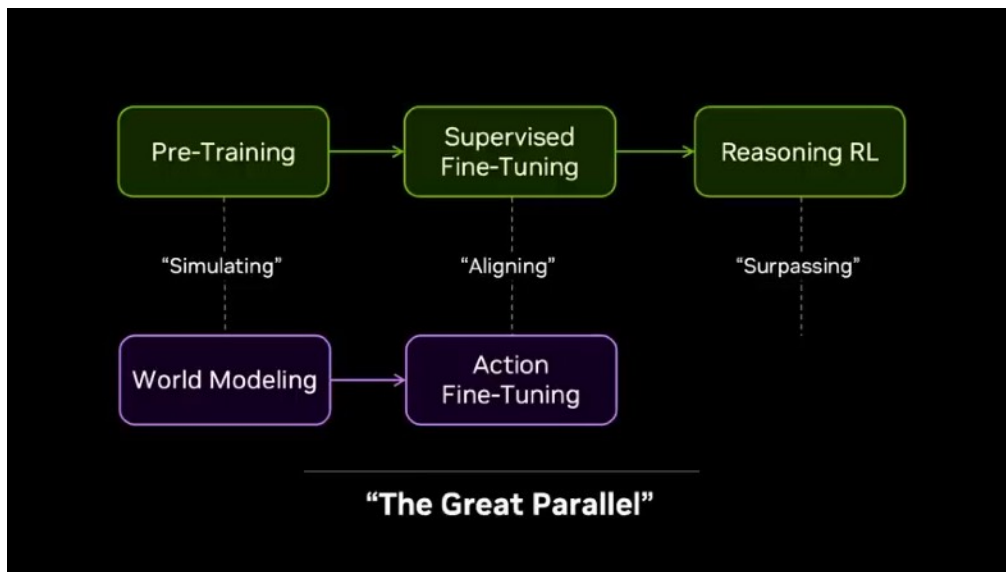
- Reasoning RL is presented as the phase that lets models go beyond imitation of human responses.
- This sets up the analogy that robotics will also need a 'surpassing' phase.



Part 2 — Foundation Models and Data

The Great Parallel: LLMs and Robotics

- This is the central thesis slide of the talk: the LLM pipeline maps onto a robotics pipeline.
- World modeling, action fine-tuning, and physical RL play roles analogous to pre-training, SFT, and reasoning RL.



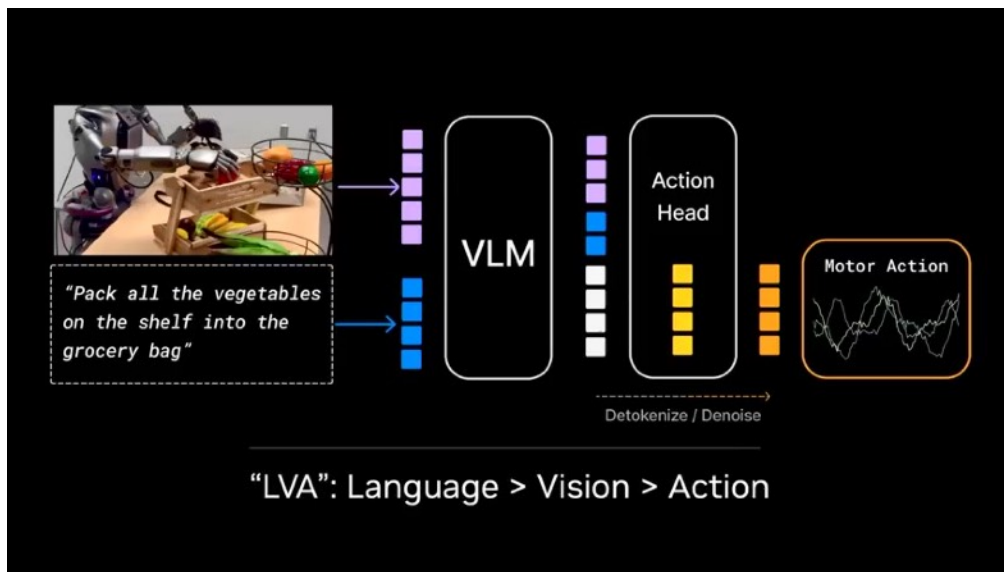
Model Strategy and Data Strategy Must Co-Evolve

- A robot foundation model needs both a scalable architecture and a scalable data engine.
- Good models fail on weak data, and abundant data fails if the model cannot use it.



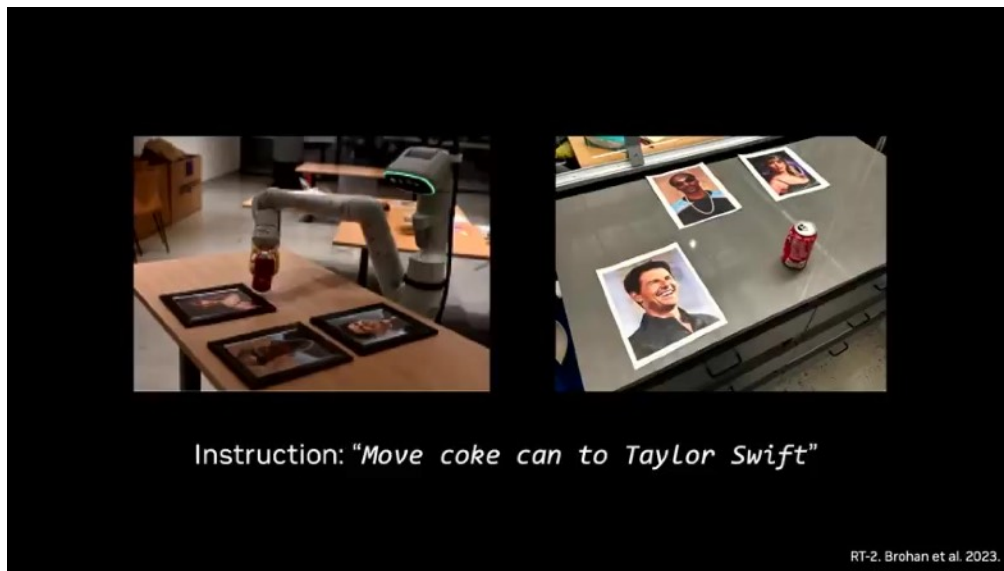
VLA as the Basic Robotics Backbone

- The Vision-Language-Action stack uses language and visual understanding to generate motor actions.
- This is the modern successor to older task-specific robot policies.



RT-2: Semantic Knowledge Transfers into Action

- RT-2 shows that internet-scale VLM pre-training can transfer semantic knowledge into manipulation.
- The robot can ground concepts it never saw during robot-only training.



Video Is a Second Pre-Training Paradigm

- Jim argues that large-scale video pre-training can teach robots physical regularities, not just language semantics.
- Even noisy internet video may still contain useful supervision about dynamics and affordances.



Video Models Learn Physics-Like Structure

- Physics-aware video generation suggests that models can internalize geometry, motion, and contact statistics.
- This supports the idea of pre-training robotics systems with world-model objectives.



But Video Models Still Fail at Long-Horizon Reasoning

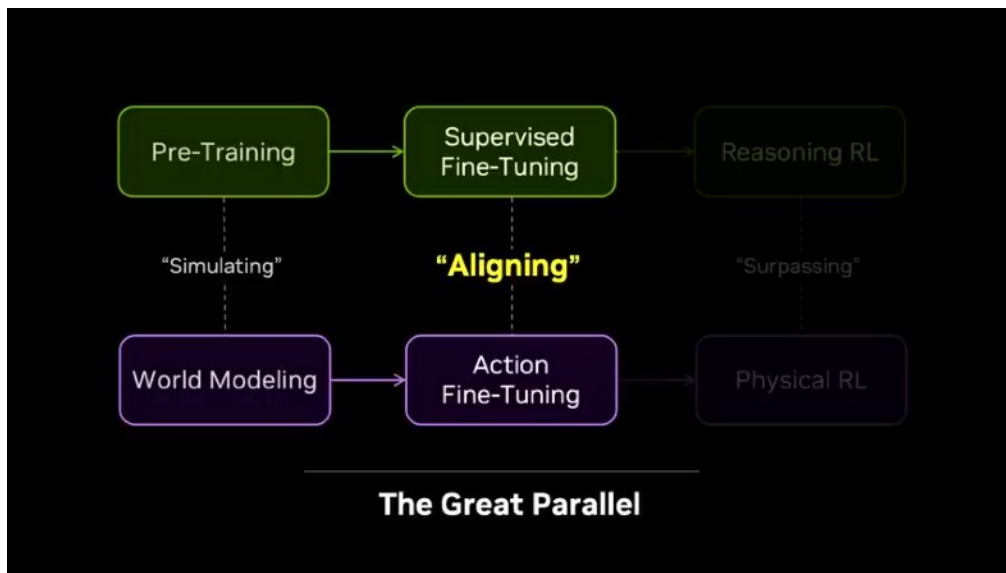
- Jim explicitly separates short-term physical realism from long-horizon decision making.
- This motivates the need for a second phase beyond pure video pre-training.



Part 3 — Action Fine-Tuning and World Models

Action Fine-Tuning Aligns a World Model to a Robot Body

- Action fine-tuning is the robotics analog of instruction tuning: it adapts a broad model to an embodiment.
- It connects a general world model to concrete motor outputs and task execution.



Zero-Shot and One-Shot Skill Transfer

- The DreamZero and Franka demos illustrate that action fine-tuning can produce nontrivial generalization to unseen tasks.
- The point is not one specific robot, but the emergence of reusable control competence.



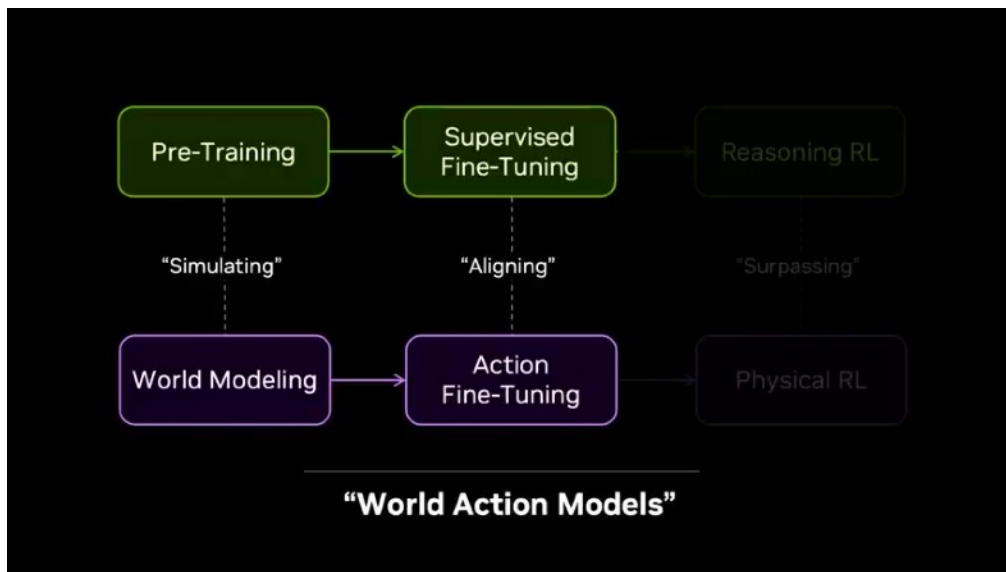
World Models Support Internal Simulation

- The 'world model view' slide suggests that robots can plan through internal rollouts rather than direct reflex alone.
- This is the conceptual step from VLA-style action mapping to deeper world-action reasoning.



World Action Models (WAM) as the Full Robotics Recipe

- Jim reframes the modern stack as a World Action Model rather than a plain VLA.
- The emphasis moves from static perception-to-action mapping toward predictive embodied modeling.



'VLA Is Dead, Long Live WAM'

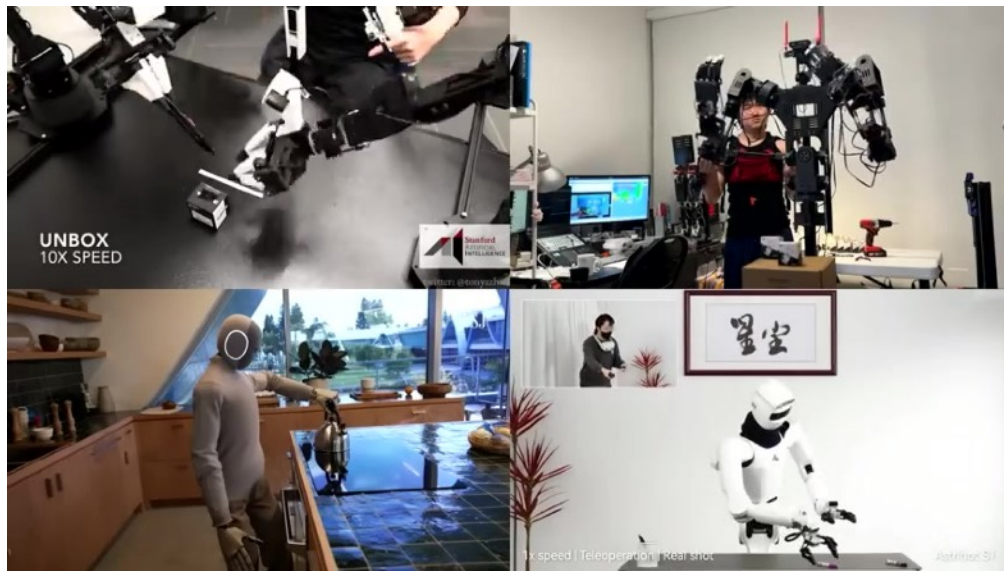
- The meme slide is provocative, but the underlying argument is architectural.
- He is claiming that action alone is not enough; world modeling has to become first-class.



Part 4 — Data Engines

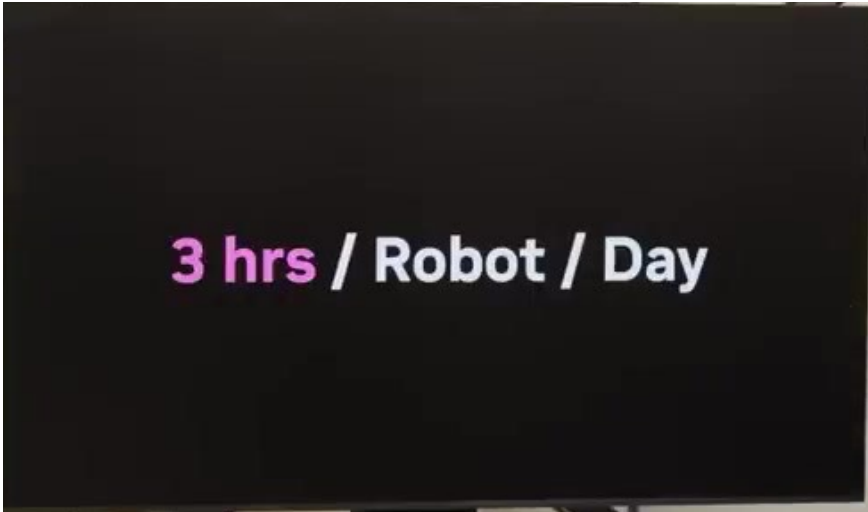
Teleoperation Became the Default Data Engine

- Teleoperation dominated because it provided embodiment-aligned data with task labels and action traces.
- This made it the most practical way to bootstrap large robot behavior datasets.



Teleoperation Has a Hard Throughput Ceiling

- Jim argues that teleop is fundamentally bounded by human labor, robot access, and operator fatigue.
- This makes it too slow to be the sole data engine for robot foundation models.



3 hrs / Robot / Day

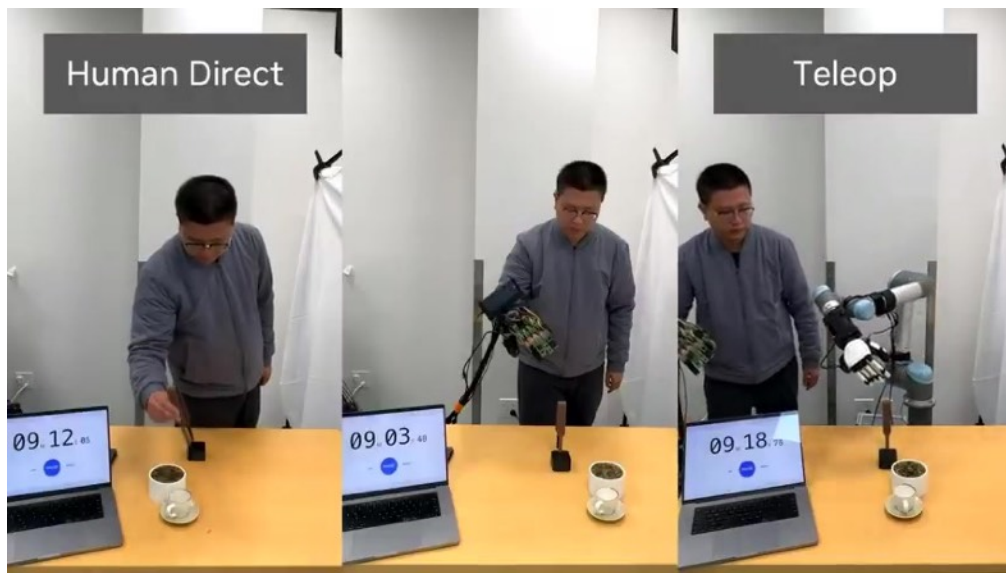
Lower-Friction Interfaces Try to Fix the Data Bottleneck

- Handheld tools, data gloves, and dexterous interfaces all try to preserve action alignment while lowering collection cost.
- These systems are transitional technologies on the way to more scalable pipelines.



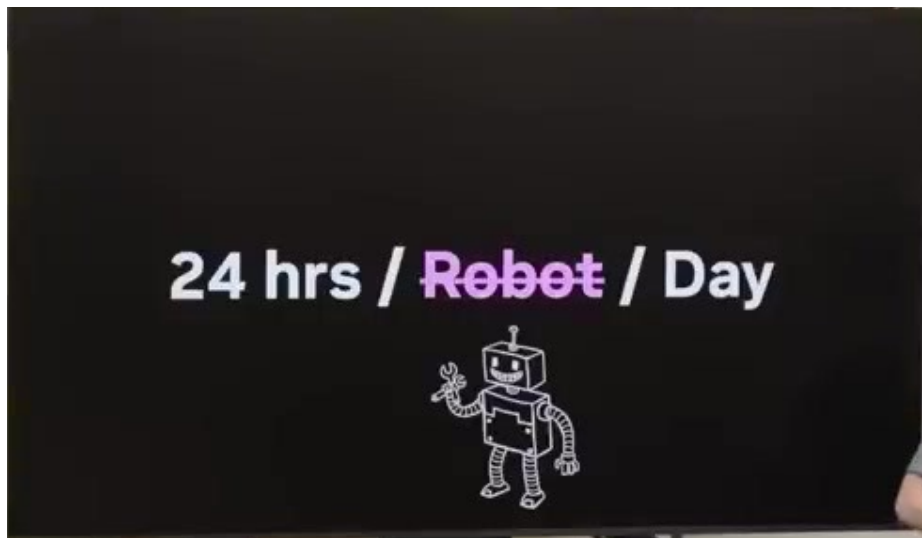
Human-Level Demonstration Speed Matters

- A core metric is whether a data interface lets the operator move at roughly natural human speed.
- If the interface is too slow, scaling collapses even if the data is high quality.



Autonomous Rollouts Turn a Robot into Its Own Data Generator

- Once a policy becomes competent, the robot can produce new trajectories far faster than teleoperation alone.
- This is the beginning of a flywheel: trained policy leads to autonomous data, which improves the next policy.



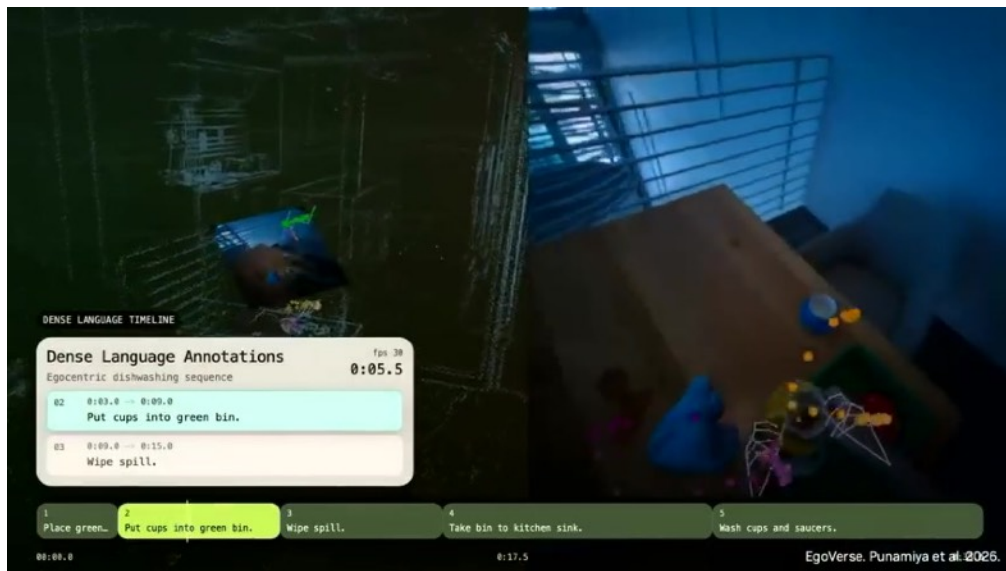
Ego-Video Is the Internet-Scale Substrate for Embodied Learning

- First-person human video is attractive because it is abundant and behaviorally rich.
- It offers a way to pre-train on physical interaction before robot-specific alignment.



EgoVerse Turns Raw Ego-Video into Structured Supervision

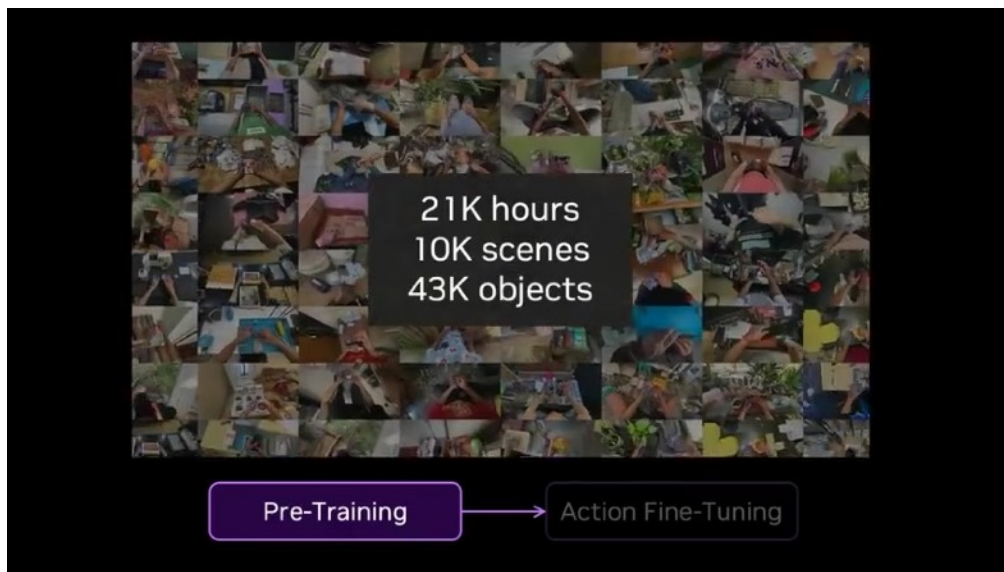
- Dense language annotation makes ego-video more useful for pre-training and temporal grounding.
- This connects internet-scale passive data to embodied downstream training.



Part 5 — Physical RL and Simulation

GR00T Demos: Foundation Models Begin to Transfer Across Tasks

- The GR00T demonstrations are used as evidence that a single model can cover multiple dexterous behaviors.
- They anchor Jim's claim that robot foundation models are no longer purely speculative.



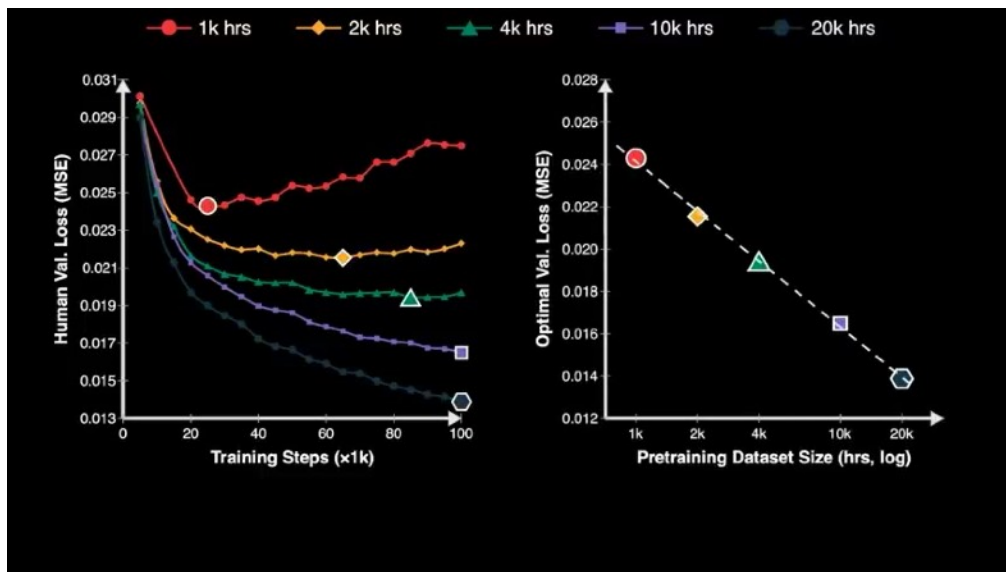
Scale Comes from Pre-Training Data plus Embodiment-Specific Fine-Tuning

- Jim separates broad pre-training on huge datasets from the smaller action fine-tuning set tied to a specific robot.
- This mirrors the way large language models separate general pre-training from downstream alignment.



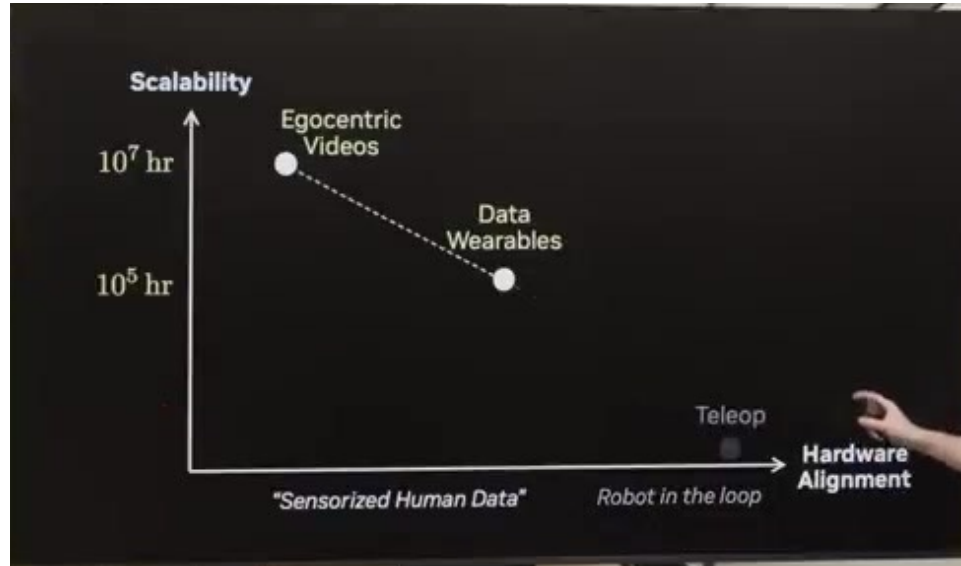
Scaling Laws Already Appear in Robot Pre-Training

- The charts argue that more pre-training data produces consistent loss reductions in robotics as well.
- This is an important bridge from anecdotal demos to scaling-law style evidence.



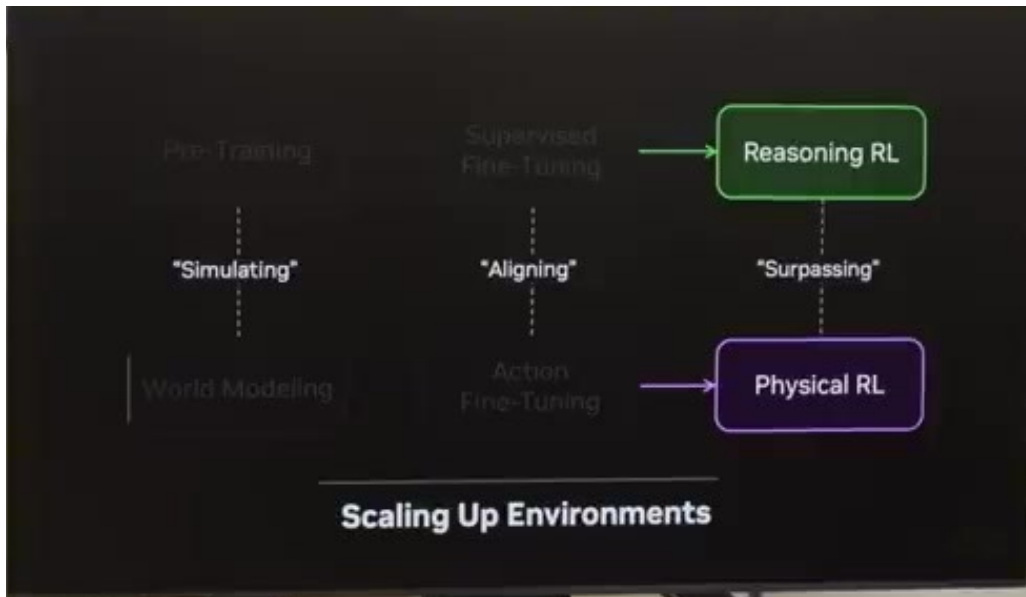
Scalability and Hardware Alignment Trade Off Against Each Other

- Teleop is highly aligned but scarce, while ego-video is abundant but poorly aligned to robot actuation.
- Sensorized human data is presented as the middle ground that may unlock the next scale jump.



Physical RL Is the Robotics Version of the Surpassing Phase

- Just as reasoning RL aims to exceed human demonstrations in LLMs, physical RL aims to exceed teleoperated trajectories in robotics.
- This is the third phase in Jim's robotics pipeline.



Physical RL in the Real World Is Expensive but Powerful

- The assembly example shows why physical RL is attractive for fine motor behaviors that require repeated improvement.
- The limitation is that real-world training is expensive, slow, and physically constrained.



Digital Twins and Digital Cousins Expand the Training Universe

- Real2Sim2Real increases scale by turning one environment into many trainable variants.
- Digital cousins matter because exact replicas are not enough; variation is what drives robustness.



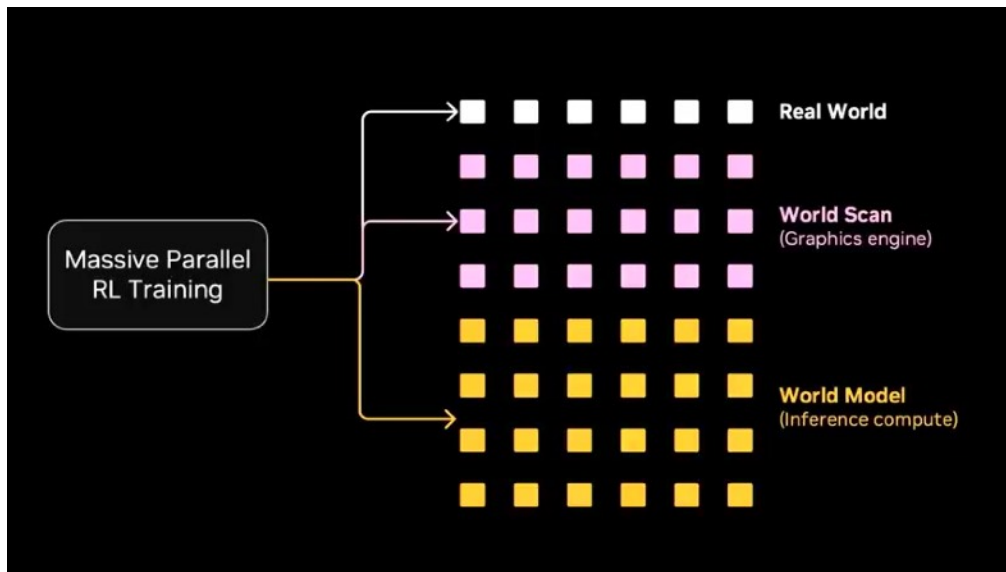
World Models Become Simulators

- Jim's later examples suggest that world models may eventually serve as interactive simulators for training control.
- This reduces the dependence on hand-authored simulators and graphics pipelines.



Massive Parallel RL Needs Three Inputs

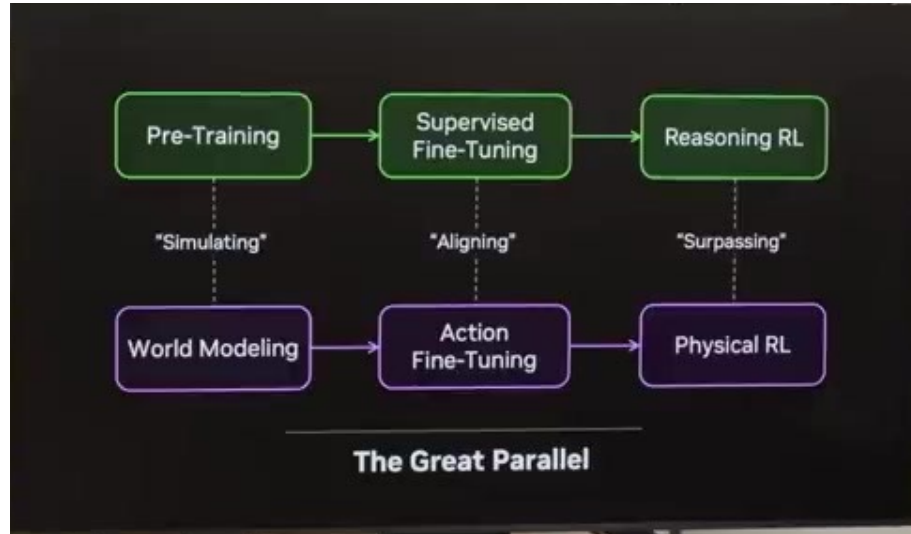
- Jim's summary diagram combines real-world data, scanned worlds, and model-generated worlds.
- The scaling vision is not one simulator, but a blended training ecosystem.



Part 6 — The Robotics Endgame

One Recipe Now Connects LLMs and Robotics

- At this point Jim returns to the full parallel to show that the analogy is complete, not partial.
- Robotics is positioned as the next domain to inherit the scaling structure of modern AI.



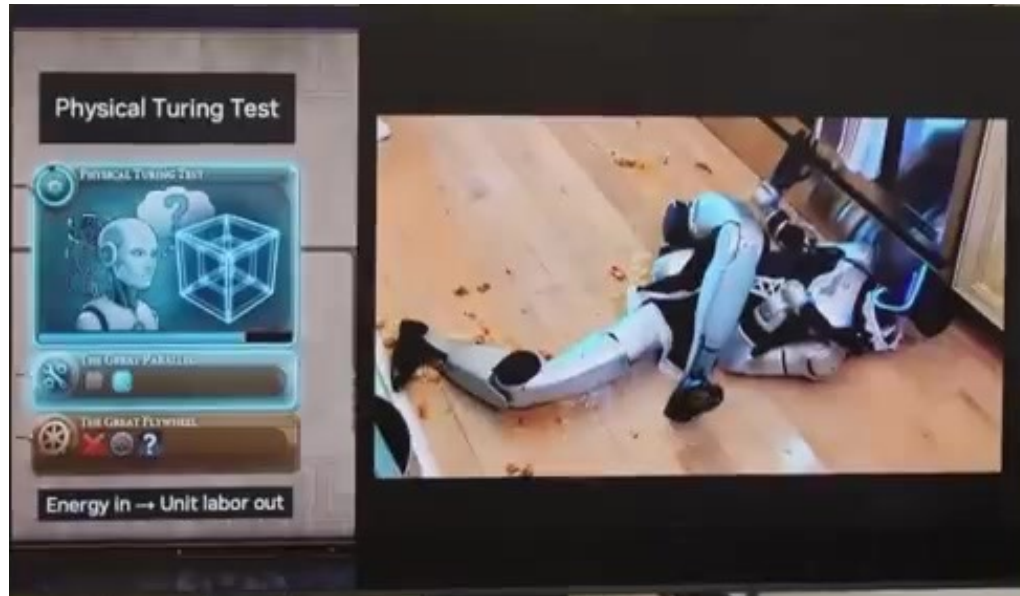
The Tech Tree Metaphor

- Jim uses a civilization-style tech tree to turn robotics progress into a sequence of unlockable capabilities.
- This metaphor helps the audience see today's models as intermediate technologies, not the endpoint.



The Near-Term Milestone Is a Physical Turing Test

- The first milestone is not general-purpose autonomy everywhere, but human-comparable labor on bounded tasks.
- The failure demo underscores that today's systems are still far from that threshold.



The Timeline from AlexNet to General-Purpose Robotics

- Jim situates the current moment as a midpoint between the start of deep vision and a possible 2040 robotics horizon.
- The rhetorical move is to treat robotics today like deep learning circa its earlier breakout era.

